Supplementary Material S1

Search Strings by Platform

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

1 Summary of execution

The three main searches (Scopus, Web of Science Core Collection and IEEE Xplore) and the complementary search in ACM Digital Library were executed on 15 February 2026. The four strings share the same Boolean structure and descriptors, organised into three conceptual blocks joined by AND: population (ASD), concept (facial emotional expressiveness) and method (machine learning). The differences across platforms are field syntax and temporal restriction. The three main searches were updated on 12 September 2026; the update did not identify additional eligible studies.

Platform	**Temporal range**	**Records retrieved**	**Execution date**
Scopus	2019–2025	935	15/02/2026
Web of Science Core	2018–2026	942	15/02/2026
IEEE Xplore	2018–2026	38	15/02/2026
ACM Digital Library	2018–2026	5	15/02/2026
Total		**1,920**	

Distribution by platform at each stage of the selection process:

Stage	**Scopus**	**WoS**	**IEEE**	**ACM**	**Total**
Identification (records retrieved)	935	942	38	5	1,920
Removed before screening	293	684	25	5	1,007
Screening (after removal)	642	258	13	0	913
Full-text retrieval	60	17	8	0	85
Passing full-text screening	25	7	2	0	34

The 5 ACM records were excluded before screening: 2 were unavailable under open access and 3 were subject to institutional access restrictions at the time of the search; all failed CI(4) and were removed in the pre-screening step. They do not contribute to the screened count.

Of the 34 studies passing full-text screening, 4 were excluded following the post-search editorial status verification (Section 2.7 of the manuscript). Of the remaining 30, 21 constitute the eligible corpus and 9 are documented as contextual evidence outside the eligible set (Section 2.4).

Forward and backward citation chasing identified five candidate studies; their full-text screening was completed during the September 2026 update and none met the eligibility criteria, so this procedure did not modify the eligible corpus.

The Scopus string was restricted to 2019–2025, whereas the Web of Science and IEEE Xplore strings applied 2018–2026. The aggregate coverage of the three main searches spans the full 2018–2026 conceptual window: Web of Science contributed the two eligible studies published

in 2018 and IEEE Xplore the single one published in 2026, and the 25 studies retrieved in Scopus fall within 2019–2025. The years 2020 and 2023 appear with zero eligible studies; both columns are retained in the corresponding table of the manuscript to preserve the continuity of the temporal series.

2 Conceptual blocks common to the four platforms

Block	Descriptors joined by OR
Population	autism; ASD; autism spectrum; autistic
Concept	emotion; facial expression; affective; social communication
Method	machine learning; deep learning; artificial intelligence; computer vision; neural network; classification; recognition

The three blocks are combined with AND. Filters applied across all four platforms: document type restricted to journal article and conference paper; open access; and English language.

3 String executed in Scopus

Temporal range: 2019–2025. Records retrieved: 935.

```
TITLE-ABS-KEY(("autism" OR "ASD" OR "autism spectrum" OR "autistic") AND ("emotion" OR "facial expression" OR "affective" OR "social communication") AND ("machine learning" OR "deep learning" OR "artificial intelligence" OR "computer vision" OR "neural network" OR "classification" OR "recognition")) AND PUBYEAR > 2018 AND PUBYEAR < 2026 AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp")) AND (LIMIT-TO(OA,"all")) AND (LIMIT-TO(LANGUAGE,"English"))
```

TITLE-ABS-KEY searches title, abstract and keywords simultaneously. The filter `PUBYEAR > 2018 AND PUBYEAR < 2026` is exclusive at both ends and therefore covers 2019–2025, as confirmed by the year distribution of the 935 exported records (`data/scopus-data.csv`).

4 String executed in Web of Science Core Collection

Temporal range: 2018–2026. Records retrieved: 942.

```
TS=(("autism" OR "ASD" OR "autism spectrum" OR "autistic") AND ("emotion" OR "facial expression" OR "affective" OR "social communication") AND ("machine learning" OR "deep learning" OR "artificial intelligence" OR "computer vision" OR "neural network" OR "classification" OR "recognition")) AND PY=(2018-2026) AND DT=(Article OR Proceedings Paper) AND LA=(English)
```

TS= (Topic) is the functional equivalent of TITLE-ABS-KEY in Scopus. PY= restricts by year, DT= by document type and LA= by language. Open access was applied as interface refinement.

5 String executed in IEEE Xplore Digital Library

Temporal range: 2018–2026. Records retrieved: 38.

```
(("All Metadata":"autism" OR "All Metadata":"ASD" OR "All Metadata":"autism spectrum" OR "All Metadata":"autistic") AND ("All Metadata":"emotion" OR "All Metadata":"facial
```

```
expression" OR "All Metadata":"affective" OR "All Metadata":"social communication")
AND ("All Metadata":"machine learning" OR "All Metadata":"deep learning" OR "All
Metadata":"artificial intelligence" OR "All Metadata":"computer vision" OR "All
Metadata":"neural network" OR "All Metadata":"classification" OR "All
Metadata":"recognition"))
```

Executed through Command Search. Temporal restriction (2018–2026), document type (Journals and Conferences), open access and English language were applied as interface refinements. All Metadata is broader than TITLE-ABS-KEY and additionally includes IEEE indexing terms.

6 String executed in ACM Digital Library

Temporal range: 2018–2026. Records retrieved: 5.

```
[[All: "autism"] OR [All: "ASD"] OR [All: "autism spectrum"] OR [All: "autistic"]]
AND [[All: "emotion"] OR [All: "facial expression"] OR [All: "affective"] OR [All:
"social communication"]] AND [[All: "machine learning"] OR [All: "deep learning"] OR
[All: "artificial intelligence"] OR [All: "computer vision"] OR [All: "neural
network"] OR [All: "classification"] OR [All: "recognition"]]
```

Executed through ACM DL Advanced Search. Temporal restriction (01/2018–12/2026), open access and English language were applied as interface refinements. The [All: "term"] operator searches across all indexed metadata fields. All 5 retrieved records failed CI(4): 2 were unavailable under open access and 3 were subject to institutional access restrictions at the time of the search. None proceeded to screening.

7 Pilot searches

Pilot searches with therapeutic and gamification terminology were executed to verify coverage:

- serious game AND autism AND emotion recognition
- therapeutic monitoring AND ASD AND machine learning
- mHealth AND autism AND facial expression

These jointly retrieved fewer than 2% of additional non-overlapping records, confirming that the main string adequately captures the therapeutic monitoring literature in these databases. The limitation is documented in Section 2.9 of the manuscript.

8 Complementary procedures

- Forward and backward citation chasing: five candidate studies identified. Full-text screening completed during the September 2026 update; none of the five met the eligibility criteria in full.

Procedures not executed: formal PRESS review by external specialist librarian; platform similarity recommendation functions; systematic exploration of grey literature.

Search update, 12 September 2026. The three main searches were re-executed on 12 September 2026 with the same strings and eligibility criteria. The citation chasing candidates were also screened to completion at this date. Neither the database update nor citation chasing yielded additional eligible studies. The eligible corpus comprises N = 21 studies after the eligibility verification documented in Section 2.4 of the manuscript.